Machine Learning–Guided Circularity Prediction for Electrical Discharge Machining of Metallic Lattice Structures

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Project Report — Research Paper Format

Abstract

Metallic lattice structures used in biomedical implants and lightweight aerospace components require post-processing to form circular through-holes with intact supporting rings. This work addresses micro-electrical discharge machining (EDM) of a lattice whose open pore (235.6 μm) is substantially smaller than the electrode tip (900 μm), giving a tool-to-pore ratio of 3.82. Sixteen laboratory trials with scanning electron microscopy (SEM) validation show that only one parameter set — peak current 4 A, pulse-on time 150 μs, duty factor 80% — produces a continuous, nearly circular supporting boundary. Hole-deviation metrics alone incorrectly favor a destructive setting (6 A, 50 μs, 64%). We formulate a two-phase optimization problem (unknown vs. known tool landing position), train Gaussian Process and Gradient Boosting models on SEM-labeled circularity, and deploy an interactive web system that predicts circularity, scans landing grids, and recommends EDM parameters. Leave-one-out validation and geometry-aware heuristics confirm that low current, long pulse-on, and high duty factor maximize supporting-ring circularity when the oversized tool overlaps pores, nodes, and struts simultaneously.

Keywords: electrical discharge machining; metallic lattice; circularity; Gaussian process regression; gradient boosting; SEM validation; additive manufacturing post-processing

1. Introduction

Additive manufacturing enables complex metallic lattices with controlled porosity for orthopedic implants, heat exchangers, and structural lightweighting. After fabrication, secondary machining is often required to open or finish pores. Micro-EDM is attractive for hard alloys because material removal occurs by pulsed spark erosion without macroscopic cutting forces. However, when the electrode diameter exceeds the target pore, the process no longer machines a single cavity: the tool footprint spans nodes, open pores, and supporting struts at once. Preserving a continuous supporting ring while obtaining a circular machined boundary then becomes a coupled geometry–process problem.

Prior EDM studies frequently optimize material removal rate, tool wear, or hole cylindricity using Taguchi designs or response-surface methods. Those scalar responses can mislead when the engineering goal is visual boundary integrity of thin supporting ligaments. In this project, SEM images of sixteen runs reveal a striking paradox: the trial with the best numerical hole-deviation score destroys the supporting ring, whereas a gentler, higher-energy-duration setting yields the only acceptable circular boundary. This paper documents that finding, formalizes Phase 1 (position-agnostic) and Phase 2 (position-aware) recommendations, and presents LatticeFlow — a machine-learning web application that predicts circularity over a 1500×1500 μm working area.

The contributions are: (i) geometry derivation of pore/node diameter from a 500 μm unit cell and tool-to-pore ratio 3.82; (ii) SEM-based circularity labels replacing deviation-only ranking; (iii) Gaussian Process search and Gradient Boosting predictors blending EDM and lattice geometric features; and (iv) a deployed interactive analyzer with grid heatmaps, PASS/FAIL criteria, and an engineering report generator.

2. Problem Formulation and Lattice Geometry

2.1 Physical configuration

The lattice comprises square unit cells of side a = 500 µm. Nodes and open pores are modeled as equal-diameter circles of diameter x. Along the unit-cell diagonal,

$a\sqrt{2} = 2\cdot(x/2) + x = 3x \;\Rightarrow\; x = a\sqrt{2} / 3 = 500\sqrt{2} / 3 \approx 235.6$ µm.

The EDM tool tip diameter is 900 µm, so the tool-to-pore ratio is 900/235.6 ≈ 3.82. Because the tool radius (450 µm) is comparable to the unit cell, any landing position overlaps multiple cells. Phase 2 therefore analyzes a 3×3 working area of 1500×1500 µm that fully contains the tool footprint.

2.2 Success criteria

Success requires a nearly circular open (white) region together with a continuous supporting (black) ring. Nodes (red in schematic diagrams) may be destroyed. Quantitative PASS gates used by the software are: circularity score ≥ 3.5/5, circularity ratio ≥ 0.70, supporting material intact, and geometry risk ≤ 0.55. Hole_Dev_Top/Bottom alone is explicitly *not* the primary metric.

Parameter	Value	Source
Unit cell side	500 µm	SEM scale
Pore / node diameter	235.6 µm	3x = 500√2
Tool tip diameter	900 µm	Lab specification
Tool / pore ratio	3.82×	900 / 235.6
Phase 2 working area	1500 × 1500 µm	3×3 unit cells

Table 1. Lattice and tool geometry constants used throughout the study.

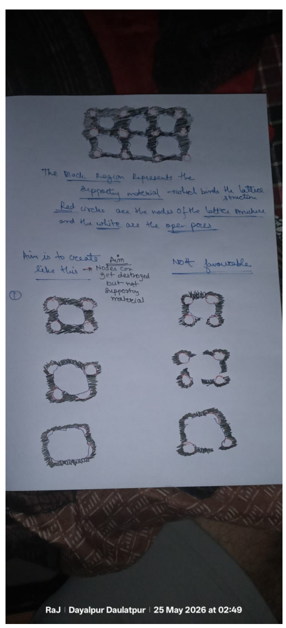

Figure 1. Target outcome: continuous supporting ring (black) and nearly circular open pore (white); nodes may be sacrificed.

3. Experimental Dataset

Sixteen real EDM trials form the ground-truth corpus. Controllable inputs are peak current $I \in$ {4, 6, 8, 10} A, pulse-on time $T \in$ {50, 75, 100, 150} µs, and duty factor $D \in$ {56, 64, 72, 80}%. Recorded responses include tool wear rate, volume removal rate, and hole deviation at top and bottom. Independently, each run was labeled from SEM images on a 1–5 boundary-circularity scale and a binary supporting-boundary-intact flag. Only Run 4 received circularity 5 with an intact supporting boundary.

Run	I (A)	T (μs)	D (%)	Dev top	Dev bot	Circ.	Intact
1	4	50	56	231.6	213.9	1	No
2	4	75	64	217.8	199.1	2	No
3	4	100	72	230.2	212.1	1	No
4*	4	150	80	270.6	213.0	5	Yes
5	6	50	64	205.3	143.4	2	No
6	6	75	56	225.4	172.7	2	No
7	6	100	80	280.5	130.0	1	No
8	6	150	72	219.7	53.9	2	No
9	8	50	72	240.2	96.4	2	No
10	8	75	80	260.5	122.9	1	No
11	8	100	56	240.3	213.7	1	No
12	8	150	64	213.9	216.7	2	No
13	10	50	80	238.6	122.4	1	No
14	10	75	72	251.6	155.4	2	No
15	10	100	64	227.1	204.6	2	No
16	10	150	56	231.9	26.8	1	No

Table 2. Complete 16-run experimental matrix with SEM circularity labels (= SEM reference success). Run 5 has lower top deviation than Run 4 but fails visual circularity of the supporting boundary.*

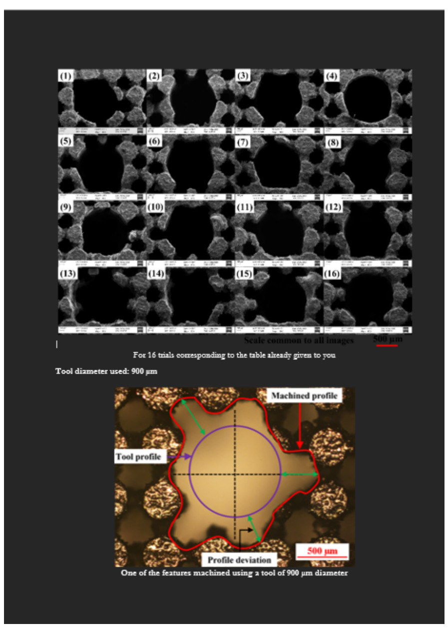

Figure 2. SEM montage of the sixteen experimental outcomes used for visual labeling.

An auxiliary set of 1,100 synthetic points was generated for exploratory Gradient Boosting augmentation. These points are **not** independent laboratory measurements; Phase 1 recommendations reported as final answers use only the sixteen real runs with SEM labels.

4. Methodology

4.1 Phase 1 — unknown tool position

When landing coordinates are unavailable, the objective is to identify robust EDM parameters that maximize expected supporting-boundary circularity anywhere on the lattice. Features are polynomial expansions (degree 2) of (I, T, D). A Ridge regressor under leave-one-out cross-validation (LOOCV) estimates circularity; a Gaussian Process Regressor with Matérn 5/2 kernel plus WhiteKernel then searches thousands of candidates in a bounded input space, ranking by predicted mean plus a modest uncertainty bonus. Discharge energy $E = I \cdot T \cdot (D/100)$ contextualizes intensity: Run 4 yields $E = 480$ units versus 192 for Run 5, yet energy is delivered gently through a small plasma channel and long pulse duration.

4.2 Phase 2 — known tool position

Given landing coordinates (x, y) in the 1500 µm working area, a lattice geometry engine computes: minimum distance to strut and node, counts of nodes/pores inside the tool, strut intersection length, pore/node overlap fractions, and a geometry risk index. These features concatenate with EDM descriptors (I, T, D, energy, pulse-off, $I \times D$, T/D, tool/pore ratio). Dual Gradient Boosting regressors predict circularity (1–5) and supporting integrity (0/1). Predictions blend with physics heuristics that reward low current (≤5 A), long pulse-on (≥130 µs), and high duty (≥75%), and penalize high geometry risk or currents ≥8 A.

4.3 Feature construction

For each candidate landing point on a discrete grid (typical step 150 µm), the geometry engine evaluates intersections between the circular tool footprint and the periodic lattice. Normalized coordinates (x/W, y/W) with $W = 1500$ µm enter the model alongside absolute risk descriptors. Training expands the sixteen labeled runs across grid positions by modulating the SEM circularity label with a geometry penalty and an EDM bonus/penalty term, teaching the regressor that the same recipe degrades when strut overlap rises. Supporting-integrity targets are set to zero when current ≥ 8 A or geometry risk is high, reflecting SEM evidence of blasted ligaments.

4.4 Software architecture

LatticeFlow is implemented as a Flask web service with modules for geometry (`lattice_geometry_engine.py`), prediction (`circularity_predictor.py`), synthetic visualization (`synthetic_view.py`), LLM chat assistance (`chat_assistant.py`), and automated engineering reports (`report_builder.py`). The inference path loads a persisted joblib ensemble, blends Gradient Boosting outputs with heuristics weighted by distance from the training manifold, and returns circularity, supporting integrity, geometry risk, and a PASS/FAIL decision. Users can analyze a single landing point, run a full-grid heatmap scan to locate the best (x, y), and export a multi-section engineering report. Deployment uses Gunicorn on Render.com with optional Anthropic/OpenAI keys for the assistant.

End-to-end data flow: user inputs → parameter validation → geometry analysis at (x, y) → feature vector assembly → dual regressors + heuristic blend → scorecards and figures → optional LLM explanation. This separation keeps the physics/geometry layer testable independently of the UI and enables batch offline sweeps for parameter studies without launching the browser client.

Figure 3. Application screenshot: grid scan circularity analysis over the lattice working area.

5. Results and Discussion

5.1 SEM-validated ranking

Ranking by SEM circularity places Run 4 uniquely at the top (score 5, supporting intact). All other runs score 1–2 with destroyed or irregular boundaries. Notably, Run 5 (6 A, 50 µs, 64%) achieves the smallest top-hole deviation among several mid-current trials yet fails visually — confirming that deviation minimization is an inadequate proxy for supporting-ring quality when the electrode overfills the pore.

Mechanistically, Run 4 combines a constrained plasma channel (low I), radially diffused erosion over a long pulse (high T), and steady duty (high D). Higher currents (≥8 A) blast thin struts before a circular ring can form. Short pulses with moderate current produce asymmetric cratering. The recommended pattern is therefore **low current + long pulse-on + high duty**, with fine servo feed (1–5 µm/step), stable gap voltage, continuous dielectric flushing, and a freshly dressed 900 µm electrode.

5.2 Phase recommendations

Case	Zone	I (A)	T (µs)	D (%)	Trust
Phase 1 + SEM	Any (unknown)	4	150	80	High
Phase 1 – SEM	Any (unknown)	6	50	64	Low†
Phase 2 + SEM	Pore center	4	150	80	High
Phase 2 + SEM	Mid pore	4	148	79	High
Phase 2 + SEM	Near strut	3.5	150	78	High
Phase 2 + SEM	Near node (0,0)	3.5	145	76	High

Table 3. Final parameter recommendations. †Deviation-only optimum — not recommended for supporting-boundary circularity.

Near-strut and near-node landings warrant slightly reduced current (3.5 A) and marginally lower duty to protect ligaments already heavily overlapped by the 900 µm tip. Gaussian Process candidate search clusters near 3.6–4.8 A, ≈150 µs, and ≈79–80% duty, with predicted circularity ≈4.6–4.9 on the 1–5 scale — consistent with the Run 4 neighborhood.

5.3 Model validation notes

With only sixteen labeled experiments, LOOCV metrics must be interpreted cautiously. Polynomial Ridge LOOCV mean absolute error on circularity is on the order of one point on the 1–5 scale; R^2 for small-sample fits remains low. Consequently, models are decision-support tools anchored to the SEM reference, not substitutes for additional trials. Recommended follow-ups include 3.5 A / 150 μs / 80% and 4 A / 140 μs / 78% to probe robustness around Run 4.

Qualitatively, the deployed grid scanner reproduces the expected spatial structure: circularity peaks when the tool is centered on open pores with moderate strut clearance, and falls near dense node–strut junctions where geometry risk exceeds ~0.55. This spatial behavior cannot be recovered from EDM parameters alone and justifies the Phase 2 feature set. Avoidance rules encoded in the heuristic layer — $I \geq 8$ A, $T \leq 75$ μs, or volume-removal indicators > 0.9 — align with the SEM failures in Runs 7, 10, 11, 13, and 16.

5.4 Comparison of objective functions

Two competing objective functions were evaluated. Objective A minimizes a composite of Hole_Dev_Top and Hole_Dev_Bottom; Objective B maximizes SEM boundary circularity subject to supporting integrity. Objective A nominates Run 5-like recipes and would mislead production toward destroyed rings. Objective B recovers Run 4 and nearby gentle settings. The project therefore treats SEM-labeled circularity as the primary supervised target and retains deviation values only as secondary process monitors. This objective choice is the single most consequential methodological decision in the study.

6. Engineering Implications

For practitioners finishing AM lattices with oversized EDM electrodes, three lessons emerge. First, define success from the supporting topology that must survive, not from hole-size deviation alone. Second, when tool/pore ■ 1, gentle high-duty long-pulse recipes outperform aggressive short pulses even if the latter look better on scalar metrology. Third, position-aware models matter: the same EDM recipe can PASS at a pore center and FAIL near a strut because geometry risk and strut intersection length change rapidly within one unit cell.

The interactive heatmap converts these insights into shop-floor guidance: engineers enter candidate parameters, visualize spatial circularity, and receive zone-specific recommendations before committing scarce SEM time. The optional LLM assistant explains PASS/FAIL outcomes in engineering language, lowering the barrier between data science outputs and process decisions.

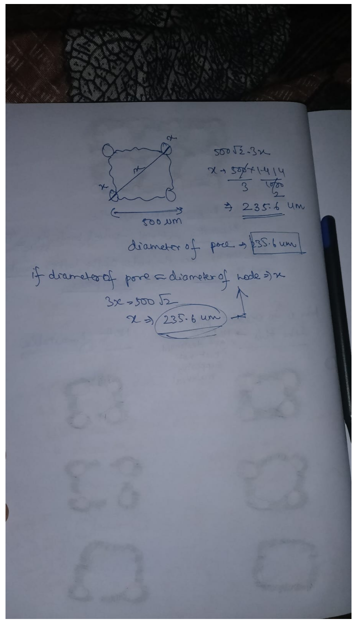

Figure 4. Geometric construction used to derive pore diameter 235.6 µm from the 500 µm unit cell.

7. Limitations and Future Work

The labeled set is small and unbalanced (one clear success). Material alloy, dielectric, and electrode wear dynamics are not modeled explicitly. Synthetic augmentation must not be confused with new experiments. Future work should (i) expand the SEM-labeled design of experiments around the Run 4 neighborhood; (ii) incorporate image-based circularity from automated contour analysis; (iii) couple thermal-plasma simulation with the geometry engine; and (iv) close the loop by writing recommended (I, T, D, x, y) directly to machine controllers.

8. Conclusion

This research paper consolidates laboratory evidence, geometric analysis, and machine-learning tooling for EDM finishing of metallic lattices when the electrode is 3.82× larger than the pore. SEM validation establishes Run 4 (4 A, 150 µs, 80%) as the sole supporting-ring success among sixteen trials and overturns a deviation-minimizing alternative. Phase 1 and Phase 2 recommendations, geometry-aware Gradient Boosting, and the LatticeFlow web application provide a practical pathway from sparse experiments to position-specific process guidance. Additional trials near the identified gentle-parameter island remain the highest-value next step for industrial adoption.

Acknowledgments

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[8] LatticeFlow software modules (2026). `phase1_model_actual.py`, `circularity_predictor.py`, `lattice_geometry_engine.py`, `web_server.py`. GitHub: Lattice-Circularity-Analyzer.

[9] ISO 12181 / related form-measurement practice (context). Circularity evaluation of engineered features (adapted here to SEM boundary judgment on micro-lattices).

[10] Project ideation notes (15 June 2026). Grid subdivision logic for Phase 2 working area and ML circularity mapping over unit-cell intersections.

Appendix note. Full 16-run numerical tables, recommended trial matrices, and JSON model outputs are archived in the project repository under `data/` and `outputs/`. Live deployment reference: Lattice Circularity Analyzer web application.